

Polymarket Baseline Pipeline: One-Page App Summary

What it is

A Python data pipeline plus Streamlit discovery app for Polymarket market analysis. It ingests market metadata and token price history, then builds normalized datasets, analysis outputs, and optional research artifacts.

Who it's for

Primary user/persona (explicit): Not found in repo. Inferred from CLI/docs and modules: quant/data researchers and analysts studying Polymarket market behavior.

What it does

- Discovers active (and optional resolved) events/markets from Gamma API.
- Extracts event, market, condition, and token asset IDs into normalized tables.
- Fetches CLOB price history with token-bucket rate limiting, retries, and chunked windows.
- Writes raw JSONL/parquet plus upserted parquet datasets for repeatable runs.
- Computes features, quality gates, clustering, and report/plot artifacts.
- Optionally ingests orderbooks and volume, then runs signals, backtests, and candidate ranking.
- Provides a Streamlit UI for filtering bets, inspecting price curves, and reviewing map/tag views.

How it works (repo-evidenced architecture)

- Entry points: ``polymarket-pipeline`` CLI (``run`` and ``resolve``) and Streamlit ``discovery_ui.py``.
- Service clients: ``PolymarketClients`` wraps Gamma/CLOB/Data APIs with token-bucket pacing, retry/backoff, and optional cache.
- Pipeline flow: discover events -> normalize to events/markets/tokens -> ingest price history -> optional resolved/orderbook/volume -> analyze/cluster -> optional signals/backtest/candidates.
- Persistence and ops: parquet/json outputs under ``data/...``, run health tracking, optional GCS sync (``gcs_sync.py``) and Cloud Run scheduler scripts.

How to run (minimal)

- Install: ``python -m pip install -e .``
- Run baseline pipeline:
- ``polymarket-pipeline --output-dir data/dev --max-events 2000 --window-days 90 --interval 1h --yes-only-binary``
- Optional UI:
- ``streamlit run polymarket_pipeline/discovery_ui.py -- --data-dir data/dev``